

In remote sensing and AI, **LoRA (Low-Rank Adaptation)** is a parameter-efficient fine-tuning (PEFT) technique that allows large foundational models to be adapted to domain-specific tasks without re-training or modifying all of their parameters 1, 2.

In **SatQuery AI**, LoRA forms the core ML strategy, enabling a single open-weight Vision-Language Model (VLM) backbone to act as a versatile bank of remote-sensing specialists 1, 3.

1. How LoRA Works in SatQuery AI

Instead of fine-tuning billions of parameters across the entire model:

- **Frozen Backbone:** The heavy visual encoder (e.g., CLIP-ViT) and base LLM decoder (1–4B parameters) remain completely frozen 2, 4, 5.
- **Low-Rank Deltas:** Small, trainable low-rank adapter matrices (rank 8–16) are attached exclusively to the LLM's attention projection layers (Q/K/V/O) 2, 5.
- **Minimal Compute:** This adapts the model to specialized remote-sensing terminology and sensor data while updating **less than 1% of the total model parameters** 2.

2. Why SatQuery AI Uses a LoRA Adapter Bank

Generic VLMs trained on broad multi-task datasets often suffer from **multi-task collapse** or task interference when fine-tuned jointly on narrow geospatial datasets 2, 6. For instance, joint multi-task training on small remote-sensing data can cause performance on specialized subtasks (like presence detection or object grounding) to drop significantly 2, 6.

To prevent this, SatQuery AI isolates task-specific knowledge into a **bank of modular LoRA adapters** built over a shared frozen backbone 1, 3, 4:

1. **adapter_vqa_caption (Joint VQA & Captioning):** Fine-tuned on BigEarthNet annotations to handle open-ended/closed-set visual question answering and scene description 2, 5, 7.
2. **adapter_grounding (Dedicated Region Grounding):** Trained separately on referring expressions to predict bounding-box token sequences 2, 5, 8. Keeping grounding in its own adapter prevents its bounding-box formatting requirements from interfering with natural-language text generation 2, 5.
3. **adapter_change_vqa (Bi-Temporal Change Analysis):** Fine-tuned on the CDVQA dataset alongside a difference-attention Change-Enhancing Module (CEM) to analyze bi-temporal image pairs 9-11.

3. Hot-Swapping & Inference Efficiency

Having separate specialist models typically requires loading multiple monolithic neural networks into GPU memory simultaneously 3, 4. SatQuery AI avoids this bottleneck through **adapter hot-swapping** 12, 13:

- The shared 1–4B backbone is loaded into GPU memory **once** at startup 4, 14.
- Each LoRA adapter delta is extremely small (**<50 MB per adapter**) and resides in GPU memory simultaneously 13.
- When the Agentic Controller routes a task (e.g., executing a compound query like "identify built-up regions, then detect changes"), model_serving applies the required LoRA delta on-the-fly per forward pass without reloading the base backbone 12, 13, 15. This keeps P95 latency low (under 3–6 seconds) while preserving high per-task accuracy 13, 16.

💡 *Would you like to examine the exact training hyper-parameters used for BigEarthNet fine-tuning, or see how adapter hot-swapping is handled inside the `model_serving` container during runtime?*